Q1. Test Connectivity Between Two Devices Using Ping

Answer:

1. Connect the laptop and a friend's phone to the same Wi-Fi network.

2. Find the IP address of both devices.

3. Open Command Prompt on the laptop.

4. Type the following command:

   bash

   ping 192.168.1.10

5. Press Enter and check the result.

Result: If replies are received, the target device is reachable. If a request times out, the device may be offline or blocking ping requests.

Q2. Run Traceroute to YouTube

Answer:

On Windows, open Command Prompt and type:

bash

tracert www.youtube.com

On Linux or Mac, type:

bash

traceroute www.youtube.com

The command shows the network hops between your device and the destination.

Result: The number of hops is the number of intermediate network steps shown before the destination is reached. Some hops may show timeouts because certain routers do not respond to traceroute probes.

Q3. Disconnect One Device and Ping Again

Answer:

- Connect both devices to the same Wi-Fi.
- Ping the target device from the laptop.
- Disconnect the target device from Wi-Fi.
- Run the ping command again.

Expected result: The command may display:

text

Request timed out.

This indicates that no reply was received within the timeout period. However, a timeout alone does not prove that the device is disconnected, because firewalls can also block ping requests.

Q4. Create a Table of Devices Connected to Wi-Fi

Answer:

On Windows, open Command Prompt and type:

bash

arp -a

This command displays IP addresses and their corresponding MAC addresses known to your computer.

No.	IP Address (Example)	Device
1	192.168.1.1	Wi-Fi Router
2	192.168.1.10	Laptop
3	192.168.1.11	Friend's Phone
4	192.168.1.12	Smart TV

Q5. One-Page Report on Network Connectivity Testing

Network Connectivity Testing Report

Objective: To test network connectivity between devices using ping and traceroute commands.

Procedure:

First, I connected my laptop and another device to the same Wi-Fi network. I checked their IP addresses using the network settings. I used the ping command to test whether the devices could communicate with each other.

Next, I ran the tracert www.youtube.com command in Windows Command Prompt to examine the network path to YouTube. The command displayed the hops between my device and the destination.

Then, I disconnected one device from Wi-Fi and repeated the ping test. I observed whether the device responded or a request timeout message appeared.

Finally, I used the arp -a command to view IP and MAC address entries and prepared a table of network devices.

Results: Ping can help check connectivity between devices. Traceroute shows the network path and hop information. The ARP command displays IP-to-MAC address mappings available to the computer.